

Raahul Datta Nidumoru

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Software Engineer with a strong foundation in backend systems, RESTful API development, and full-stack engineering, currently applying that foundation to Artificial Intelligence (AI) and Machine Learning (ML) research. Experienced in integrating Generative AI (GenAI) capabilities — including Google Gemini and OpenAI GPT-4o — into production systems, with hands-on work in Retrieval-Augmented Generation (RAG) pipelines, LLM integration, and cloud deployment (AWS, Docker, CI/CD). Growing expertise in deep learning, model evaluation, prompt engineering, and structured AI reasoning pipelines. Combines practical software engineering skills with a research-driven approach — analyzing system design, model behavior, and evaluation methodologies to build AI-driven solutions that are reliable, explainable, and evidence-grounded.

EXPERIENCE

Full Stack Developer Intern

Jan. 2026 – Apr. 2026

AD-X LIVE Private Limited

Gachibowli, Hyderabad

- Contributed to full stack development initiatives within a fast-paced team, gaining hands-on industry experience on real-world projects
- Collaborated with the development team on projects aligned with company goals and objectives
- Delivered contributions across both frontend and backend development tasks
- Gained hands-on experience applying full stack development practices in a production environment

Full Stack Developer Intern

Jun. 2026 – Aug. 2026

Redgates IT Solution

Noida, Delhi

- Contributed to the end-to-end development of a salon service booking application, working across the full stack while supporting brand and marketing initiatives
- Built and enhanced core features for a salon services booking platform, improving overall user experience
- Developed across both frontend and backend layers to deliver seamless, end-to-end functionality
- Contributed to brand identity and content strategy through blog and social media initiatives
- Designed cross-platform advertising creative for web and mobile to support the app's launch and marketing efforts

PROJECTS

Ember — AI Incident Response Engineer | *React, TypeScript, Express, PostgreSQL, Google Gemini* 2025

- Designed an AI-assisted incident investigation system that generates evidence-backed root-cause hypotheses with confidence scoring, using structured prompting over Google Gemini 2.5 Flash
- Architected a human-in-the-loop remediation pipeline enforcing mandatory approval gating before any proposed production action, with full audit-trail logging of every state transition
- Built a configurable triage-rules engine that auto-classifies incoming alerts by severity and routes them to the correct on-call team, replacing manual incident intake
- Implemented the backend with Express and Drizzle ORM over PostgreSQL, and integrated PagerDuty, Slack, and Jira APIs for end-to-end incident lifecycle automation

DNA Sequence Classification | *Python, Scikit-learn, PyTorch, NLP, Machine Learning, Deep Learning* 2026

- Developed an end-to-end DNA sequence classification pipeline for 7 gene families across Human, Chimpanzee, and Dog genomic datasets
- Implemented k -mer tokenization, CountVectorizer, Multinomial Naive Bayes, Random Forest, Decision Tree, 1D-CNN, and Transformer-based architectures
- Achieved $97.74\% \pm 0.28\%$ 5-fold cross-validation accuracy on Human sequences with Multinomial Naive Bayes and 100% zero-shot transfer accuracy on Chimpanzee sequences
- Built automated training, evaluation, inference, cross-species validation, confusion-matrix generation, and model persistence workflows with unit testing

SynapseLab | *React, TypeScript, Node.js, Express, Google Gemini, Socket.IO* 2026

- Built an AI-powered startup validation platform that transforms business ideas into structured market intelligence and actionable execution strategies using Google Gemini
- Developed an end-to-end analysis workflow covering market viability, growth potential, competitor benchmarking, SWOT analysis, funding opportunities, market trends, and execution roadmaps
- Implemented real-time collaborative analysis rooms using Socket.IO and designed a modular React/TypeScript frontend with an Express backend
- Integrated persistent analysis workflows and export capabilities for generating PDF, CSV, and PNG reports

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, SQL
AI/ML: Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Generative AI (GenAI), Large Language Models (LLM), Google Gemini, Claude, TensorFlow, PyTorch, Scikit-learn, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG)
Frameworks & APIs: FastAPI, Spring Boot, React, Node.js, Express.js, RESTful APIs
Databases: PostgreSQL, MySQL, MongoDB
Cloud & DevOps: AWS (EC2), Docker, Git, CI/CD, GitHub Actions
Tools & Libraries: Pandas, NumPy

EDUCATION

K L Deemed to be University <i>Bachelor of Technology in Computer Science - 8.21 CGPA</i>	Hyderabad, Telangana <i>Aug. 2023 – May 2027</i>
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